# Operators in PQL

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This topic gives you an overview of the arithmetic and boolean operators that you can use in PQL expressions, and the precedence of all operators.

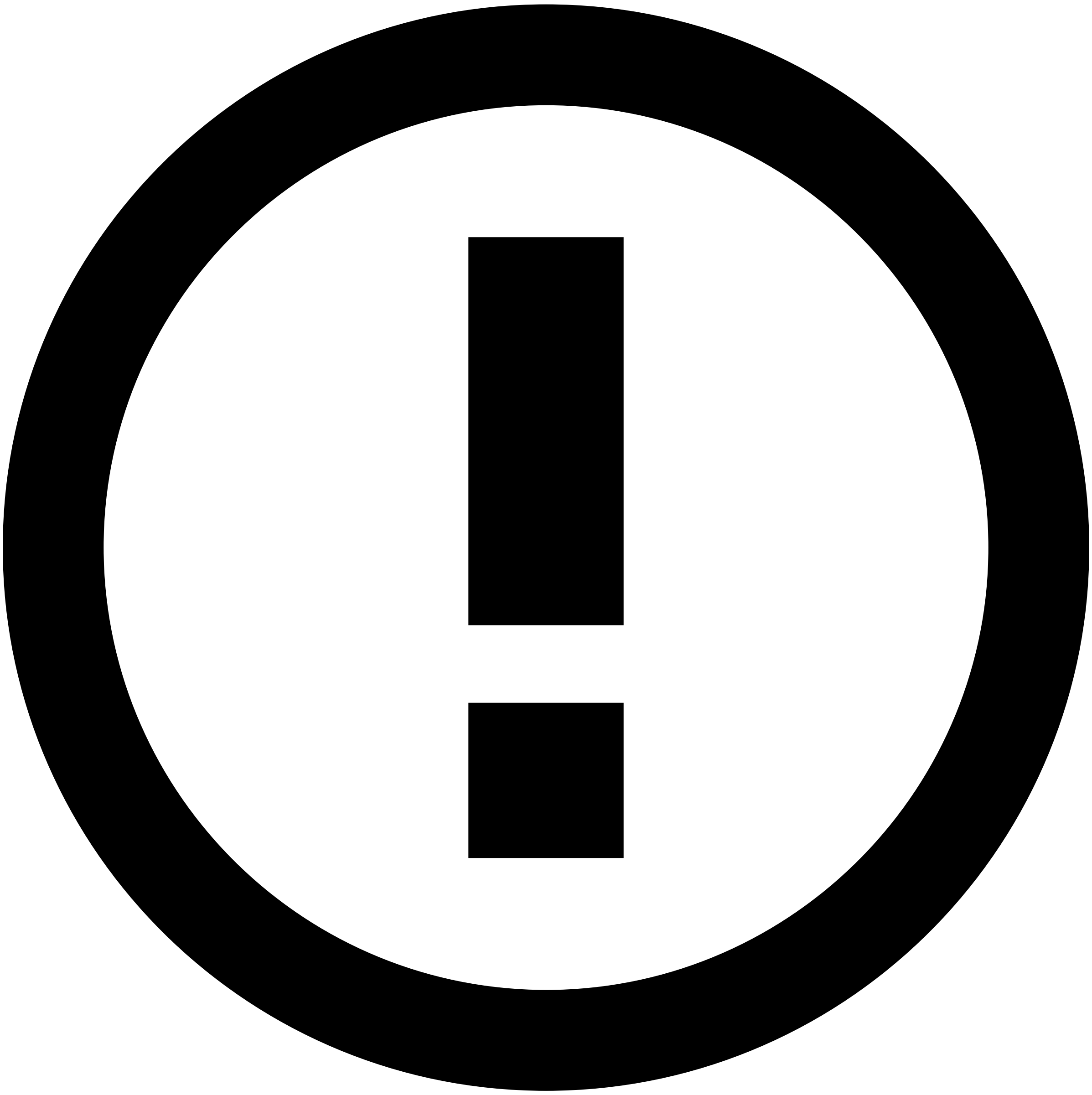
## Arithmetic operatorsâ

PQL uses the following arithmetic operators:

| Operator | Meaning |
| --- | --- |
| `+` | addition |

| Operator | Meaning |
|---|---|
| `-` | subtraction |
| `*` | multiplication |
| `/` | division |
| `%` | integer remainder |
| `^` | exponentiation |

Example: `Balance - Amount`



Type promotion when using arithmetic operators

The arithmetic operators obey the following type promotion rules:

1. If at least one of the arguments is a double, the result is a double.
2. If at least one of the arguments is a float, the result is a float, unless the result is a double because of the first rule.

3. If at least one of the arguments is a long, the result is a long, unless the result is a double or float because of the first two rules.

## Boolean operatorsâ

Boolean operators are operators that return a boolean value. They include:

- `<` , `<=` , `==` , `!=` , `>=` , `>`

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Only numeric values can be compared with the <, <=, >=, and > operators.

- `and` , `or` , `!` (not)

Example:

```
POSEntryMode == 'Swiped' or POSEntryMode == 'Keyed' or POSEntryMode == 'Chip'
```

## Operator precedenceâ

The following table lists PQL operators according to their precedence order. Operators at the top of the table have higher precedence.

| Operator type | Operator(s) |
|---|---|
| grouping | `(expr)` |
| unary | `+expr` , `-expr` , `!` |
| arithmetic - exponential | `^` |
| cast | `exp:type` |
| arithmetic âmultiplicative | `*` , `/` , `%` |
| arithmetic âadditive | `+` , `-` |
| coalesce | `??`<br><br>Note that this operator has a different precedence in RDL. |
| comparison | `<` , `<=` , `==` , `>=` , `>` , `!=` |
| logical âAND | `and` |
| logical âOR | `or` |
| assignment | `=` |

Cast operator precedence

Notice that the *cast* operator precedence lies between arithmetic operators. If an expression needs to be cast before an exponentiation, it needs to be enclosed in parentheses (*grouping* operator).

The following rules apply:

- operators with higher precedence than other operators are evaluated before them;
- operators on the same line have the same precedence:
    - all binary operators except for the assignment operator are evaluated from left to right;
    - assignment operator is evaluated right to left.

## Examplesâ

Arithmetic and boolean operators of PQL follow the common rules of mathematics regarding functionality and precedence.

Arithmetic and boolean operators have higher precedence than the `??` (coalesce) operator, meaning that the `??` operator gets evaluated after arithmetic and boolean operators.

Example: `a + b ?? c + d` is equivalent to `(a+b) ?? (c+d)`.